MILITARY SPECIFICATION

PROJECTILE 155MM: XM864 METAL PARTS FOR BASE BURNER ASSEMBLY

This specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality evaluation provisions, and packing for Base Burner Assembly Metal Parts for the XM864 projectile.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures for Sampling by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKING DRAWINGS

8597488	Plug
8822518	Cup
9381110	Retainer
9381116	Housing, Igniter

(Copies of specifications, standards, handbooks, drawings, and publications required by contractor's in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Defect standards. Defect standards must be approved by the technical engineering agency. All visual standards for defects will be established for each contract after the first months production or whenever requested by the contracting activity.

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3.3 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.4 Control dimensions.

3.4.1 Tool control dimensions. Dimensions marked tool control shall be gaged after eight hours maximum production on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimension and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance and to institute corrective action that assures drawing requirements are being met throughout the life of the tool.

3.4.2 Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use. Dimensions to be gaged for process control are to be assured to a .40% AQL.

3.5 Protective finish. Details of the proposed procedure for protective coating the various components shall be submitted in writing to the Government Technical Agency and written approval received prior to initiation of production. The proposed procedure shall include materials, process and equipment to be used by the contractor, detailed method of control including limits for time, temperature, concentration and all other information pertinent to the process. Approval of process, materials, or equipment implies no guarantee of acceptance of the results obtained in use.

3.6 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Burr and sharp edges. No component shall have a burr or sharp edge which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.6.2 Foreign matter. All parts and assemblies shall be free of chips, dirt, grease, oil, rust, or other foreign matter which may interfere with final inspection and completely free of same prior to shipping.

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3.6.3 Threads. Threads shall be full and undamaged, for the entire minimum length or depth as required by the applicable drawings.

3.6.4 Painting. Painting shall be complete, void free, and undamaged.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements and for the provisions of MIL-A-48078 as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the quantities as indicated.

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PART DESCRIPTION	DRAWINGS	QUANTITY
Cup	8822518	25
Housing, Igniter	9381116	10
Plug	8597488	10
Retainer	9381110	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70674 (AR)

PARAGRAPH	TITLE: Metal Parts for Base Burner Assembly 155MM: XM864		SHEET 1 OF 1		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup (Dwg. 8822518) Examination for defects	25	-	3.1	4.4.2.1
	Retainer (Dwg 9381110) Examination for defects	10	-	3.1	4.4.2.2
	Plug (Dwg 8597488) Examination for defects	10	-	3.1	4.4.2.3
	Housing, Igniter (Dwg 9381116) Examination for Defects	10	-	3.1	4.4.2.4

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 or MIL-STD-1235. Inspection Level shall be II.

b. Automatic gaging. Facilities equipped with government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with the contract.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70674 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8822518
4.4.2.1	Cup				NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, min.		.40%	3.1	Gage
102	Outside diameter, min.		.40%	3.1	Gage
Minor					
201	Inside diameter		.65%	3.1	Gage
202	Thickness thru bottom		.65%	3.1	Gage
203	Protective finish missing or incomplete		.65%	3.1	Visual
204	Evidence of poor workmanship		.65%	3.6	Visual
NOTES:					

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70674 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Retainer		9381110
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.1	Gage
102	Thread pitch diameter		0.40%	3.1	Gage
103	Thread major diameter		0.40%	3.1	Gage
104	Largest inside diameter		0.40%	3.1	Gage
105	Smallest inside diameter		0.40%	3.1	Gage
106	Depth of largest inside diameter		0.40%	3.1	Gage
107	Largest outside diameter		0.40%	3.1	Gage
108	Smallest outside diameter		0.40%	3.1	Gage
109	Length from front face to largest outside diameter		0.40%	3.1	Gage
110	Depth of slots (2 places)		0.40%	3.1	Gage
111	Width of slots		0.40%	3.1	Gage
112	Length of thread pilot		0.40%	3.1	Gage
113	Distance between slots		0.40%	3.1	Gage
114	True position of slots		0.40%	3.1	Gage
Minor					
201	Width of thread relief		0.65%	3.1	Gage
202	Diameter of thread pilot		0.65%	3.1	Gage
203	Diameter of thread relief		0.65%	3.1	Gage
204	Protective finish missing or incomplete		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Plug				8597488
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Smallest inside diameter		.40%	3.1	Gage
102	Largest inside diameter		.40%	3.1	Gage
103	Pitch diameter		.40%	3.1	Gage
104	Major diameter		.40%	3.1	Gage
105	Distance between flats		.40%	3.1	Gage
106	Depth of largest inside diameter		.40%	3.1	Gage
107	Height of crown		.40%	3.1	Gage
108	Length from bottom to top of cylindrical outside diameter		.40%	3.1	Gage
Minor					
201	True position of smallest inside diameter		.65%	3.1	Gage
202	Cylindrical outside diameter		.65%	3.1	Gage
203	Length from bottom to bottom of flats (2 places)		.65%	3.1	Gage
204	Protective finish missing or incomplete		.65%	3.1	Visual
205	Evidence of poor workmanship		.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70674 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.4	Housing, Igniter		9381116
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Base thread pitch diameter		0.40%	3.1	Gage
102	Base thread minor diameter		0.40%	3.1	Gage
103	Small thread pitch diameter		0.40%	3.1	Gage
104	Small thread minor diameter		0.40%	3.1	Gage
105	Total depth from front face		0.40%	3.1	Gage
106	Smallest inside diameter		0.40%	3.1	Gage
107	Depth from front face to end of base thread relief		0.40%	3.1	Gage
108	Depth of slot		0.40%	3.1	Gage
109	Depth from front face to start of smallest inside diameter		0.40%	3.1	Gage
110	Diameter of top row of port holes (random one hole)		0.40%	3.1	Gage
111	True position of slot		0.40%	3.1	Gage
112	True position of second largest inside diameter		0.40%	3.1	Gage
113	Any hole incomplete or blocked		0.40%	3.1	Gage
114	Diameter of middle row of port holes (random one hole)		0.40%	3.1	Gage
115	Diameter of bottom row of port holes (random one hole)		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70674 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	Housing, Igniter				9381116
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Total number of port holes		0.65%	3.1	Visual
202	Location of port holes		0.65%	3.1	Gage
203	Second smallest outside diameter		0.65%	3.1	Gage
204	Second largest outside diameter		0.65%	3.1	Gage
205	Depth of base thread pilot		0.65%	3.1	Gage
206	Diameter of base thread pilot		0.65%	3.1	Gage
207	Total length		0.65%	3.1	Gage
208	Width of base thread relief		0.65%	3.1	Gage
209	Smallest outside diameter		0.65%	3.1	Gage
210	Largest outside diameter		0.65%	3.1	Gage
211	Diameter of base thread relief		0.65%	3.1	Gage
212	Second largest inside diameter		0.65%	3.1	Gage
213	Length of largest outside diameter		0.65%	3.1	Gage
214	Length from front face to end of second largest inside diameter		0.65%	3.1	Gage
215	Length of smallest outside diameter		0.65%	3.1	Gage
216	Small thread relief diameter		0.65%	3.1	Gage
217	Depth from front face to rear of second largest inside diameter		0.65%	3.1	Gage
218	Width of small thread relief		0.65%	3.1	Gage
219	Protective finish missing or incomplete		0.65%	3.1	Visual
220	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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4.4.3 Testing. Testing is described in the Quality Conformance Inspection Tables and herein.

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

5. PACKAGING

5.1 Domestic shipment plug and cup. Packaging, packing and marking for the plug and for the cup shall be in accordance with MIL-STD-1169, Level B.

5.2 Domestic shipment retainer and igniter housing. Packaging, packing and marking for the retainer and for the igniter housing shall be in accordance with MIL-STD-1169, Level B. In addition, packaging shall be, as a minimum, single item packaging as stated in the paragraph entitled "Arrangements" of MIL-STD-1169 and moisture proof as stated in the paragraph entitled "Item Protection" of MIL-STD-1169. Additional packaging shall be provided as required to prevent wearing of the oxide coating during shipment.

6. NOTES

6.1 Intended use. The base assemblies covered by this specification are intended for the Projectile, 155MM: XM864.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements indentified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement Part 27, Sub-Part 27.410-6 (DD Form 1423) are

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invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph	Data Requirement Title	Applicable DID
4.4.4	Inspection Equipment Descriptive Documentation	DI-R-1714
3.5	Procedure Report	* DI-T-5372 (Tailored)

* Delete a, b, c, d, of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits. (Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.8 Subject term (keyword) listing.

Ammunition
Artillery
Military Specification
Base Burner Parts
Turnip
Munition
Projectile, 155MM, XM864, Metal Parts For Base Burner Assembly

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

Custodian:
Army - AR

Preparing activity
Army - AR

(Project 1320-A944)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70674	PROJECTILE 155MM: XM864 METAL PARTS FOR BASE BURNER ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

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